3 Months Learning Journey (2026)

> **Department:** SDD | **Number of Internships:** 2 | **Stack:** Angular / Java / MSSQL Server / PostgreSQL
> **Core Portals:** `iBanking` (Customer Internet Banking) & `Adminconsole` (Back-Office Admin Console)

Schedule

No.	Week	Assignment	Document
Month 1 — Foundation & Environment Setup			
1	**Week 1**	**Onboarding & Environment Setup:** Install development tools (Node.js, Angular CLI, JDK 17,/ VS Code, Git, / SSMS, Postgres/ Postman). Clone iBanking_portal and adminconsole repositories along with backend services. Configure local environments, run both portals locally, and understand multi-project repository structure.	**Weekly Report**
2	**Week 2**	**Learning & Discussion: MFS Dual-Portal Architecture Overview:** Study architecture differences between customer-facing iBanking_portal and back-office adminconsole. Understand authentication flows (Customer OTP/JWT vs. Admin RBAC & session management), API gateway routing, and Serverless service communication.	**Weekly Report**
3	**Week 3**	**Learning & Discussion: Angular Fundamentals in Banking Portals:** Deep-dive into Angular components, reactive forms, custom validators (account number format, amount limits), routing guards (AuthGuard, RoleGuard), HTTP interceptors (token injection, error modal handler), and RxJS observables in both portals.	**Weekly Report**
4	**Week 4**	**Hands-on: Read & Trace Existing End-to-End Flow:** Trace a complete financial transaction flow across both portals: Customer initiates Fund Transfer / Bill Payment on iBanking_portal (UI Form → Validation → Service → REST API → DB transaction) and staff views transaction record / audit log in adminconsole.	**Weekly Report**
Month 2 — Core Development Skills			
5	**Week 5**	**Learning & Discussion: Java Serverless REST API for Financial Services:** Controller, Service, and Repository layers; DTO pattern, validation (@Valid), error handling (@ControllerAdvice), idempotency key handling for transactions, and Maker-Checker approval workflow design.	**Weekly Report**
6	**Week 6**	**Learning & Discussion: MFS Database Design & Ledger Concepts:** Core schemas across customer and admin modules (Customer, Account, Transaction, TransferLimit, FeeConfig, AdminUser, RolePermission, AuditLog). Dr/Cr logic, ACID transactions, and indexing.	**Weekly Report**

No.	Week	Assignment	Document
7	**Week 7**	**Hands-on: Bug Fix Assignment:** Fix 2–3 assigned minor bugs across iBanking_portal and adminconsole (e.g., input masking/validation for account numbers, date-picker filter issue on Admin transaction history, error toast handling on failed API calls). Submit via Git branch & Merge Request (MR).	**Weekly Report + MR Link**
8	**Week 8**	**Hands-on: Feature Enhancement:** Add a full-stack feature enhancement to an existing module (e.g., "Export Transaction Statement as PDF/CSV" with date range on iBanking_portal OR "Customer Status Lock/Unlock with Reason & Audit Log" on adminconsole). Implement frontend + backend + DB query.	**Weekly Report + MR Link**
Month 3 — Feature Development & Code Quality			
9	**Week 9**	**Learning & Discussion: Banking Security Standards & Review Process:** Financial application security best practices, OWASP Top 10 (SQL injection, XSS, CSRF), sensitive data masking (PII, Account/Card numbers), strict BigDecimal usage for monetary math, and MR review checklist.	**Weekly Report**
10	**Week 10**	**Hands-on: Mini Feature Development (Full-Stack):** Design and implement an assigned mini-feature connecting customer and admin workflows (e.g., "Beneficiary Management with OTP on iBanking_portal & Flagged Beneficiary Review on adminconsole" OR "Tiered Transfer Limit Adjustment with Maker-Checker Approval").	**Weekly Report + MR Link**
11	**Week 11**	**Hands-on: Unit Testing, Audit & Code Review:** Write unit tests for assigned services/components (JUnit 5, Mockito for backend; Jasmine/Karma for Angular), verify audit trail integrity and edge cases, participate in peer MR reviews, and document test cases.	**Weekly Report**
12	**Week 12**	**Final Project Presentation & Evaluation:** Present completed work to engineering leads and mentors (live demo of iBanking_portal & adminconsole features + architecture walkthrough), receive evaluation feedback, and submit final internship report.	**Final Report + Demo**

Deliverables Summary

Month	Key Deliverables
Month 1	Running local environment (iBanking_portal & adminconsole), Architecture documentation, End-to-end module trace report
Month 2	2–3 bug fix MRs, 1 feature enhancement MR across frontend and backend
Month 3	1 full-stack mini feature MR, Unit test coverage report & test cases, Final presentation & comprehensive internship report

Tools & Technologies

Category	Tools & Technologies
Frontend	Angular (17+/21), TypeScript, HTML5, CSS3 / Bootstrap / Angular Material
Backend	Java 17,
Database	MSSQL Server / PostgreSQL / SSMS
API & Testing	Postman, Swagger / OpenAPI/ Jasmine / Karma
Version Control	Git, GitLab / GitHub (Branching strategy, Merge Requests, Code Reviews)
IDE	Visual Studio Code (Frontend/ Backend)
Communication	Weekly Report, Daily Stand-up, MR Review, Sprint / Team Retrospective

Evaluation Criteria

Criteria	Weight	Description
Attendance & Punctuality	**10%**	Punctual attendance, active participation in daily stand-ups and team discussions
Weekly Report Quality	**20%**	Thoroughness, clarity, and timely submission of weekly progress and learning reports
Code Quality (MR Reviews)	**30%**	Adherence to coding standards, clean architecture, security awareness, and test coverage
Problem-solving & Initiative	**20%**	Analytical thinking, debugging capability, proactiveness in learning, and asking right questions
Final Presentation	**20%**	Quality of live demo (iBanking_portal & adminconsole), clear slide delivery, and code defense